Trustworthy AI and Semantic Governance: A Multi-Disciplinary Framework

The accelerating integration of Artificial Intelligence (AI)—specifically Large Language Models (LLMs) and Small Language Models (SLMs)—into complex autonomous systems presents a critical engineering challenge: the transition from "Performance-First" to "Observability-First" engineering. Modern AI is inherently probabilistic, often manifesting "super-huberous" assertions (high confidence with low grounding in physical reality), which leads to hallucinations, model drift, and eventual "model collapse."

To mitigate these risks, this briefing outlines a dual-pillar strategy:

1. **The AI Circuit Breaker:** A multi-disciplinary framework that uses a "Ground Truth" deterministic veto layer to compare AI intent against sensor-derived reality.
2. **Semantic Governance:** A strategic operating model that industrializes knowledge into governable assets, using ontologies and knowledge graphs to provide "Semantic Guardrails" for agentic AI.

The ultimate goal is to move from a state of "Blind Reliance" to "Engineered Trust," ensuring that as technology advances, it remains under rigorous human and deterministic control.

The Landscape of AI Risk: Propagation of Synthetic Error

In AI-driven autonomous systems, error is not static; it evolves and amplifies through successive layers of the data-to-decision pipeline.

The Propagation Loop

- **Data Layer (Sensory Measures):** Initial errors arise from sensor noise, calibration inaccuracies, or atmospheric interference.
- **Algorithmic Layer (Deterministic Models):** Inflexible algorithms transform sensor noise into false features, passing them upstream as valid data points.
- **LLM Layer (Semantic Amplification):** LLMs attempt to find meaning in erroneous data, leading to convincing but false narratives (hallucinations).
- **Recursive Learning & Model Collapse:** When flawed actions are fed back into training data, a positive feedback loop reinforces biases. The model loses grounding in real-world physics, resulting in "model collapse."
- **Autonomous Agent (Cascading Failure):** The agent executes unintended, potentially dangerous physical actions based on a flawed internal understanding.

The AI Circuit Breaker Framework

The proposed "Ground Truth Circuit Breaker" acts as a crucial intermediary between AI reasoning and physical execution.

Core Logic and Information Flow

The framework continuously evaluates the **Delta** (discrepancy) between the AI's **Intent** (strategic planning from LLMs/SLMs) and the **Sensor State** (ground truth from deterministic sensors).

Component	Function
LLM/SLM	Provides high-level goals and tactical perception (Strategic/Tactical Intent).
Deterministic Sensors	Provides precise physical measurements (Ground Truth).
Circuit Breaker Logic	Evaluates if **Delta > Tolerance**. If true, the breaker "trips."
Response	System Halt / Safe State Activation; data associated with the violation is flagged for **Exclusion** from future recursive learning cycles.

Multi-Layer Validation Brokers

- **Knowledge Engineering:** Uses Knowledge Graphs to anchor probabilistic outputs in verifiable truths.
- **Control Theory:** Employs multi-dimensional trust metrics (robustness, transparency, fairness) to adjust tolerance thresholds.
- **Deterministic Veto:** Grants sensors absolute authority to override AI-derived intent if a direct physical threat is detected.

Semantic Governance Operating Model

Semantic Governance harmonizes meanings, ontologies, and taxonomies across an enterprise to establish a machine-readable language for business concepts.

Strategic Value Proposition

- **Single Source of Truth:** Reduces ambiguity by standardizing terms like "revenue" or "active customer" across all systems.

- **Industrialized Knowledge:** Transforms knowledge into a monetizable asset that is governable and traceable.
- **Deterministic Causality:** Maps metadata to causality, allowing stakeholders to understand *why* an AI made a specific decision.

Architectural Components

Layer	Description	Key Technologies
Knowledge Fabric	Foundational semantic modeling and high-performance storage.	Metaphactory, GraphDB
Operational Workbench	Low-code platform for monitoring, visualization, and human interaction.	Tom Sawyer Perspectives
Intelligence Layer	Supports semantic search and orchestrates AI workflows.	Vector Databases, AI Agents
Integration Backbone	Standardized, secure communication between microservices.	Model Context Protocol (MCP)

Vector Database and Context Engine Design

A high-quality agentic AI requires a context engine that integrates domain-specific ontologies with vector databases for precise retrieval.

The Five Dimensions of Relevancy

To achieve contextual relevance, the system must capture:

1. **Who:** Persona Model (user role).
2. **What:** Business Domains and Sub-Domains.
3. **When:** Time Model (past, present, future intent).
4. **Where:** Location Model.
5. **Why:** Intent/Causality Model (Diagnostic, Evaluative, Predictive, or Reasoning Chains).

Hybrid Search Strategy

The architecture utilizes a **two-level search**:

- **Level 1 (Vector DB):** Fast approximate nearest neighbor (ANN) search for semantic similarity.
- **Level 2 (Ontology):** Precise symbolic reasoning and validation conducted by domain-specific agents to ensure compliance and interpretability.

Practical Application: Financial Planning & Analysis

Applying semantic governance to the FP&A domain illustrates how qualitative policies can be mapped to quantitative metrics.

Mapping FP&A Metrics to Deontic Statements

Using the Legal Knowledge Interchange Format (LKIF), financial metrics are linked to norms:

- **Obligation:** Variance must remain under 5%.
- **Permission:** Specific ranges allowed for forecasting adjustments.
- **Prohibition:** Forbidden actions, such as misstating forecasts.
- **Violation:** Breach of KPIs or metrics outside allowed bounds.

Automated Workflows in Finance

- **Budget Forecast Creation:** FP&A Agents interact with Vector DBs and SHACL engines for policy-driven forecast generation.
- **Variance Analysis:** AI-driven root cause breakdown identifies *why* actual results deviated from plans, flagged by semantic markers in the knowledge graph.

Trustworthiness Engineering: Metrology and Metrics

Trust must be treated as a measurable physical property versus a subjective feeling.

Key Metrics for Autonomy

- **Mean Time Between Hallucinations (MTBH):** The average time a system functions before producing a high-confidence error.

- **Value-Drift Coefficient:** A numerical measure of the distance between an autonomous agent’s current objective and the original human intent.
- **Intent Delta:** The measured difference between AI logic and physical reality.

Recursive Integrity

To prevent **Model Collapse**, "tripped" data must be purged from the training queue. Only "Clean Data"—validated ground-truth successes—is permitted to update the agent's internal logic. This ensures the AI learns from its successes, not its "super-huberous" failures.

The Role of Governance and Registries

As organizations move toward agentic AI, governance must evolve from static rules to dynamic, identity-centric frameworks.

The Role of Registries

Registries act as authoritative catalogs that:

- Identify and classify all operating AI agents.
- Track the full lifecycle (development to decommissioning).
- Monitor agent behaviors and decisions through immutable audit trails.
- Enforce security via digital identities and zero-trust access controls.

Architecting for 2026

The integration of AI into critical systems necessitates a fundamental shift toward interdisciplinary collaboration. Organizations like INCOSE must lead the charge in breaking down silos between **Systems Engineering**, **Metrology**, **Knowledge Engineering**, and **Control Theory**. By building these "Ground Truth Circuit Breakers," engineers can ensure that as technology advances beyond human comprehension, it never advances beyond human control.